MAlang Lexical Analyzer

Automata Design

CS4031 — Compiler Construction | Assignment 01 | Spring 2026
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1. Regular Expressions

1.1 Keywords (Case-Sensitive, Exact Match)

Regex:
(start|finish|loop|condition|declare|output|input|function|return|break|continue|else)

Required Keywords: start, finish, loop, condition, declare, output, input, function, return, break, continue, else

1.2 Identifiers

Regex: [A-Z][a-z0-9]{0,30}

Rules: Must start with uppercase letter (A-Z), followed by lowercase letters, digits, or underscores. Maximum 31 characters total.

Valid: Count, Variable name, X, Total sum 2024

Invalid: count, Variable, 2Count, myVariable

1.3 Integer Literals

Regex: [+-]?[0-9]+ Valid: 42, +100, -567, 0

Invalid: 12.34, 1,000

1.4 Floating-Point Literals

Regex: [+-]?[0-9]+\.[0-9]{1,6}([eE][+-]?[0-9]+)?

Valid: 3.14, +2.5, -0.123456, 1.5e10, 2.0E-3

Invalid: 3., .14, 1.2345678 (¿6 decimals)

1.5 String Literals

Regex: "([^"\\\n]|\\["\\ntr])*"

Escape Sequences: \", \\, \n, \t, \r

1.6 Character Literals

Regex: ’([^’\\\n]|\\[’\\ntr])’

1.7 Boolean Literals

Regex: (true|false) (case-sensitive)

1.8 Operators

Arithmetic: (**|[+\-*/%])

+, -, *, /, %, ** (exponentiation)

Relational: (==|!=|<=|>=|<|>)

Logical: (&&|\|\||!)

Assignment: (\+=|-=|*=|/=|=)

Inc/Dec: (\+\+|--)

1.9 Punctuators

Regex: [(){}[\][,;:]]

Characters: ( ) { } [ ] , ; :

1.10 Comments

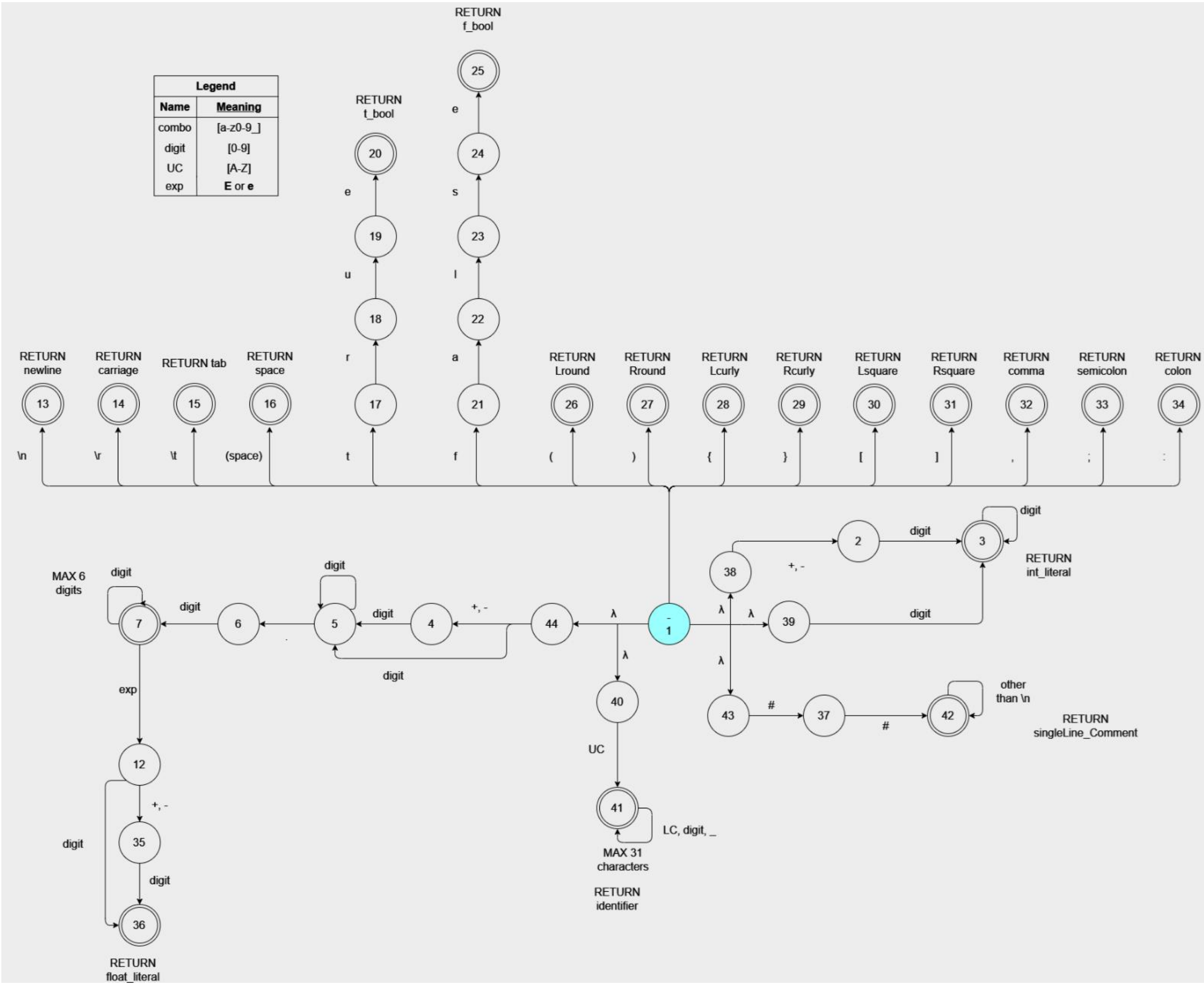
Single-line: ##[^\n]*

Multi-line: #*([^*]|*+[^*#])**+#

1.11 Whitespace

Regex: [\t\r\n]+ (skip but track line numbers)

2. NFA



Legend
Name
Meaning
combo
[a-z0-9_]
digit
[0-9]
UC
[A-Z]
exp
E or e
RETURN f_bool
RETURN t_bool
RETURN newline
RETURN carriage
RETURN tab
RETURN space
RETURN Lround
RETURN Rround
RETURN Lcurly
RETURN Rcurly
RETURN Lsquare
RETURN Rsquare
RETURN comma
RETURN semicolon
RETURN colon
RETURN int_literal
RETURN singleLine_Comment
RETURN identifier
RETURN float_literal
MAX 6 digits
MAX 31 characters
other than \n
LC, digit, _
(space)

3. Minimized DFA